Can the Existing Minas Gerais Main Grid Deliver the 2035 and 2050 Clean-Energy Portfolios?

A policy-driven transmission assessment with resource adequacy and production-cost economics

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Abstract

We test whether the existing transmission system of Minas Gerais (MG), Brazil, can deliver the generation and storage portfolios that a policy-constrained hourly capacity-expansion model selects for 2035 (state targets) and 2050 (local fossil-free supply), and which reinforcements are justified. Following the CAISO policy-driven assessment taught in the course, the portfolios are first checked for resource adequacy, then mapped to substations, tested for peak and off-peak deliverability on the EPE 2035 network model (555 monitored main-grid branches, 437 N-1 contingencies), mitigated with costed alternatives, and valued in a network-constrained 8,760-hour production-cost simulation. The 2035 portfolio is deliverable in every study condition ($\alpha \geq 5.58$) and adequate under the model's import assumption. In 2050 the midday output is deliverable ($\alpha = 1.22$), but evening delivery of the 5.3 GW battery fleet, 3.7 GW of it sited in the north, is not ($\alpha = 0.58$ to 0.62), and the network-constrained simulation sheds 689.5 GWh of load (0.71% of demand). An attributable plan of 9 reinforcement groups costing R$267–1,250 M restores N-1 deliverability (R$65.1 M more if parallel circuits share towers), cuts unserved energy to 5.9 GWh and is economic (benefit-cost ratio 44.5–208.3). Siting the same batteries at load buses removes all but one reinforcement (R$47 M) and cuts the congestion cost from R$5,824 M/yr to R$667 M/yr. Energy, not capacity, limits 2050 adequacy: a capacity-only COPT finds the system adequate, whereas an energy-limited Monte Carlo dispatch finds unserved energy in every simulated year and a dry year leaves 2,859.3 GWh unserved.

1 Question and approach

MG holds 10% of Brazil's population and plans net-zero emissions by 2050 (PLAC-MG). Hydropower supplies most of its electricity, the state is a transit corridor between the North/Northeast and the Southeast, and new solar is concentrated in the north. Our driver is the question the NREL Oregon offshore-wind study asked for its coast [12, 18]: *how much new supply can the existing system host, and does the planned portfolio need major grid improvements?*

The study follows the CAISO policy-driven assessment [10]: a base portfolio and sensitivity portfolios feed a reliability assessment, a policy-driven need assessment (peak and off-peak deliverability) and an economic assessment (production-cost simulation of congestion and curtailment). Figure 1 maps the course flow onto this study. All network work is restricted to the main grid (lines $\geq$230 kV and network transformers); 138 kV and below are outside the scope.

Data. The hourly capacity-expansion model (one weather year, 8,760 hours, MG as one area with an import bound) and its two saved solutions are the portfolio record. The network is the EPE 2035 planning case exported from ANAREDE (1,489 buses, 1,890 branches), an MG-centred equivalent whose boundary buses carry 23,616 MW of inflows and 18,002 MW of outflows. It is an evening snapshot with all 106 existing PV buses at 0 MW and is also used for 2050 (no 2050 network exists). Forced-outage data are NERC GADS class averages [17]; unit costs come from published EPE and ANEEL documents [1, 5–7]; unserved energy is valued at the 2025 Custo do Déficit [4]. Optimisation uses PyPSA [2] with linopy [8] and HiGHS [9].

2 Portfolios (1a)

The portfolio LP minimises the present value of incremental capital, fixed O&M and operating cost subject to hourly balance, policy capacity floors (PLAC-MG), resource ceilings, a monthly hydro energy budget with

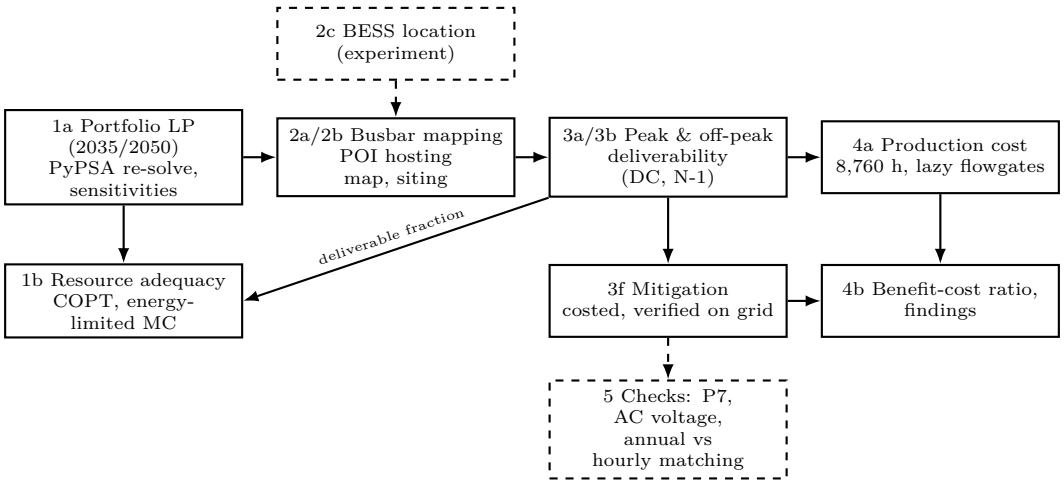


Figure 1: Study flow (solid: core assessment, after the CAISO policy-driven assessment; dashed: experiments and checks).

ramping, an import bound of 4.89 GW, and battery state-of-charge dynamics; 2050 requires local supply without imports or gas. We rebuilt the single-year LP in PyPSA (one bus; extendable generators and storage; custom constraints for the monthly hydro budget, cyclic ramping and the single battery converter). The PyPSA model reproduces the saved objectives (R$15,752.63 M and R$20,284.67 M) and capacity mixes to within 10^{-12} (relative), and a re-solve of the original code under dry hydrology (R$22,651.18 M and R$27,420.72 M) to the same precision. The model's 2035 stage counts capital and fixed O&M over 2031–2050 but operating cost only over 2031–2035, so the two objectives are not comparable; the saved portfolios are kept unchanged, since every later step depends on them. The 2035 portfolio sits entirely on the policy floors (1.0 GW central PV, 1.04 GW distributed PV, 0.88 GW biomass, 0.30 GW small hydro); the 2050 portfolio adds 12.9 GW central PV, 7.9 GW distributed PV, 2.0 GW wind, 0.4 GW biogas and 5.3 GW of 8-hour batteries (Figure 2 and Table 1). Sensitivity portfolios follow the CAISO/PJM list [16]: load ±6%, dry and (parametric) wet hydrology and gas price. In 2035 they leave the portfolio unchanged; in 2050 load +6% and dry hydrology add storage and solar.

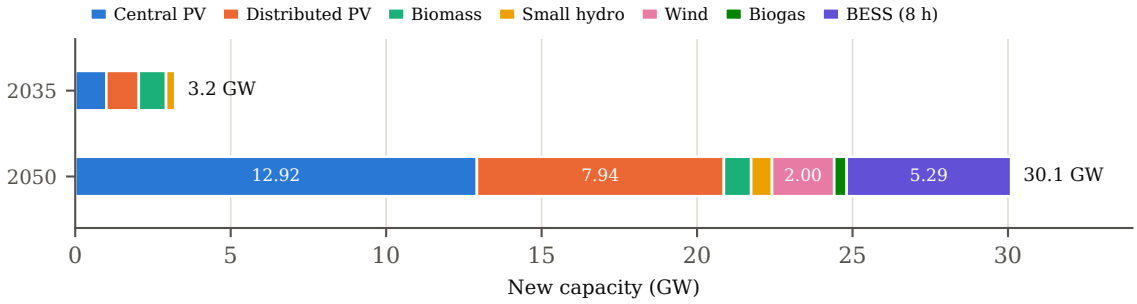


Figure 2: New capacity of the 2035 and 2050 base portfolios by technology (LP record, reproduced in PyPSA). Battery storage is shown as power. Segments too narrow for a label: 2035: central PV 1.00, distributed PV 1.04, biomass 0.88, small hydro 0.30; 2050: biomass 0.88, small hydro 0.67, biogas 0.40 GW.

3 Resource adequacy (1b)

Adequacy is tested two ways [13, 14, 20]. **Method A** is the course's capacity outage probability table: dispatchable units (reservoir hydro, gas, biogas) are convolved with GADS EFORd values (Table 2); hydro availability in each hour is the model's hourly upper bound scaled by the available share of installed units; profile-based resources and the firm import bound reduce the load; batteries shave each day's net-load peaks. LOLE, LOLH and EUE follow, and ELCC values use the CAISO calibration (add or remove perfect capacity to reach 0.1 day/yr with and without each resource class). **Method B** is a sequential Monte Carlo: two-state Markov outage histories (GADS EFORd and mean outage duration) derate hydro, gas, biogas and batteries hour by hour, and each history is an 8,760-hour minimum-cost PyPSA dispatch with the monthly hydro budgets, ramping and storage dynamics. Unit data are not in the package: the model's 13.38 GW

Table 1: Base and sensitivity portfolios (PyPSA): present-value cost, new capacity (GW) and annual energy (TWh). Sensitivities follow the CAISO/PJM list (Lectures 9–10): load ±6%, dry and wet hydrology, gas price. "Wet" is a labelled parametric +10% monthly hydro energy (no wet-year data); curtailment is not unique in the LP (zero-cost resources). Biomass + small hydro + biogas is 1.17 GW (2035) and 1.95 GW (2050) in every case.

Year	Portfolio	Cost (R$ M) ↓	PV	DG PV	Wind	BESS	Imports	Curtailed
2035	Base (LP record)	15,753	1.00	1.04	0.00	0.00	2.29	1.32
2050	Base (LP record)	20,285	12.92	7.94	2.00	5.29	0.00	15.01
2035	load +6%	17,734	1.00	1.04	0.00	0.00	5.77	0.12
2035	load -6%	15,043	1.00	1.04	0.00	0.00	1.05	3.35
2035	hydro dry	22,651	1.00	1.04	0.00	0.00	14.41	0.26
2035	hydro wet (param.)	15,110	1.00	1.04	0.00	0.00	1.16	3.27
2035	gas +50%	15,753	1.00	1.04	0.00	0.00	2.29	1.32
2035	gas -25%	15,753	1.00	1.04	0.00	0.00	2.29	1.32
2050	load +6%	24,062	17.41	7.94	2.00	6.58	0.00	16.10
2050	load -6%	17,054	10.75	7.94	2.00	3.36	0.00	17.57
2050	hydro dry	27,421	10.75	16.60	2.00	8.85	0.00	8.54
2050	hydro wet (param.)	17,469	10.75	7.94	2.00	3.74	0.00	16.60
2050	gas +50%	20,285	12.92	7.94	2.00	5.29	0.00	15.01
2050	gas -25%	20,285	12.92	7.94	2.00	5.29	0.00	15.01

of reservoir hydro is split over the 37 MG hydro-plant buses of the network in proportion to their reactive limits and then into units of at most 250 MW (125/500 MW sensitivities). Only deliverable capacity counts [20], so both methods are repeated with every new resource scaled by the peak deliverable fraction $d = \min(1, \alpha_{HSN}, \alpha_{SSN}) = 0.62$ for 2050 (Section 5).

Table 2: Forced-outage data: NERC GADS Generating Unit Statistical Brochure 4, 2016–2020, all units reporting. Mapping: reservoir hydro → Hydro 30+ MW (one plant unit below 30 MW → Hydro 1–29 MW); gas → combined cycle; biogas engines → diesel; BESS → pumped storage (no battery class in GADS). Mean outage = forced-outage hours / forced outages.

NERC GADS category	Units	Unit-years	EFORd (%)	Mean outage (h)
Hydro 30 Plus	658	2,756	4.29	119
Hydro 001-029	402	1,676	14.37	205
Combined Cycle All Sizes	345	1,420	3.87	37
Gas Turbine 50 Plus	459	2,059	9.27	72
Diesel All Sizes	151	657	7.92	35
Pumped Storage All Sizes	111	544	4.69	96

Table 3: Method A (COPT convolution, capacity only). Firm gap: perfect capacity to add (+) or that could be removed (−) for LOLE = 0.1 day/yr. "Deliverable only" scales every new resource by the peak deliverable fraction $\min(1, \alpha_{HSN}, \alpha_{SSN})$.

Year	Counting	LOLE (d/yr) ↓	LOLH (h/yr) ↓	EUE (MWh/yr) ↓	Firm gap to 0.1 d/yr (MW) ↓
2035	nameplate	0.000	0.00	0	-3,564
2035	nameplate, unit max 125 MW	0.000	0.00	0	-3,749
2035	nameplate, unit max 500 MW	0.000	0.00	0	-3,250
2035	nameplate, no firm imports	112.977	253.00	64,686	1,329
2050	nameplate	0.000	0.00	0	-1,283
2050	deliverable only	7.994	13.62	2,379	717
2050	nameplate, unit max 125 MW	0.000	0.00	0	-1,422
2050	nameplate, unit max 500 MW	0.000	0.00	0	-1,021

Findings. With the model's firm import bound, 2035 is comfortably adequate (no unserved energy in either method or hydrology; Method A could remove 3,564 MW of perfect capacity and stay at 0.1 day/yr). Without firm imports MG alone would need 1,329 MW more firm capacity. For 2050, Method A again reports an adequate system at nameplate, but Method B, which sees the monthly hydro energy budgets, finds unserved energy in every simulated year (44.0 GWh on average, P(any shortfall) = 1.00); at most 94 MW of firm capacity would bring P(any shortfall) to 0.1 (the gap search resolves 94 MW). Counting only deliverable capacity raises the Method-A gap to 717 MW, and a dry year leaves 2,859.3 GWh unserved (2.94% of load) even at nameplate, needing at most 1,594 MW of firm capacity. Method B's deliverable-only case scales the new resources' output by d in every hour and so also removes midday energy that is deliverable ($\alpha = 1.22$ in S1); its 2,863.6 GWh (normal hydrology) is a loose upper bound, against 689.5 GWh in the production-cost run that applies the

Table 4: Method B (sequential Monte Carlo, energy-limited PyPSA dispatch, 100 outage histories per case; dry = dry-hydrology analogue). EUE in MWh/yr and % of load; P(any): probability of any shortfall in the year. P(any shortfall) bounds LOLE from below (each year with a shortfall has ≥1 loss-of-load day). Firm gap: perfect capacity for P(any shortfall) ≤ 0.1, by bisection on 0–12,000 MW to 93.75 MW resolution (each value is the upper end of its interval). "Deliverable only" scales the output of new resources by d in every hour, a loose upper bound in an energy dispatch (midday output is deliverable). †Depends on how energy deficits are spread over hours (rotating 5% tranches).

Year	Hydrology	Counting	No-outage EUE ↓	EUE ↓	EUE (%) ↓	P(any) ↓	LOLE (d/yr)† ↓	Firm gap (MW) ↓
2035	normal	nameplate	0	0	0.000	0.00	0.0	0
2035	dry	nameplate	0	0	0.000	0.00	0.0	0
2050	normal	nameplate	0	43,988	0.045	1.00	28.8	94
2050	normal	deliverable only	2,800,356	2,863,617	2.947	1.00	126.8	1,688
2050	dry	nameplate	2,612,405	2,859,322	2.942	1.00	215.3	1,594
2050	dry	deliverable only	13,768,440	13,969,726	14.374	1.00	365.0	3,375

grid hour by hour (Section 7). The 2050 portfolio meets energy on average but has no reserve for outages or drought. ELCC values (Appendix Table 19) show the expected pattern: new solar contributes little at the evening peak, 8-hour batteries contribute most of their nameplate.

4 Busbar mapping and hosting capacity (2a–2c)

The reference siting places each technology's allocation in each candidate area at the nearest eligible substation (lowest eligible voltage on ties). A connection-cost MILP comparing a 4-area northern set with a 12-area regional set gives the POI half of interconnection cost (Table 5): R$1,285.1 M vs R$878.3 M in 2035 and R$5,338.2 M vs R$4,618.8 M in 2050. Network upgrade costs, the other half, are in Section 6.

Figure 3 is the per-POI hosting map in the style of NREL's Finding 1: the injection one POI can make alone before the first attributable violation. 7 of the 27 candidate stations can host at least 5 GW; the largest is Pirapora 2 (9,600 MW). 345 and 230 kV stations host 0.3–3.3 GW.

Table 5: Busbar-mapping alternatives: optimised POI connection cost (the POI half of interconnection cost; network upgrades in Section 6).

Year	Candidate set	POI connection cost (R$ M) ↓
2035	Local 4 North MG	1,285.1
2035	Regional 12 Areas	878.3
2050	Local 4 North MG	5,338.2
2050	Regional 12 Areas	4,618.8

5 Deliverability and reliability (3a–3e)

Method. Flows are incremental DC flows anchored on the AC solution, $f = F_{AC} + \mathrm{PTDF}\,\Delta p$, with line-outage distribution factors for N-1 (normal ratings for N-0, emergency ratings for N-1) on 283 lines and 272 transformer branches (437 contingencies; 44 radial outages skipped). The injected portfolio must be absorbed: following the CAISO convention all other online generation is scaled down in proportion to its output (MG hydro and thermal plus the boundary inflow equivalents, 29,253 MW of headroom), with the load at the target-year level (2050 adds 2,236 MW pro rata to existing load). A facility is attributed to the portfolio only inside the CAISO 5% circle: some study unit has a distribution factor ≥5% on it or a flow impact ≥5% of its rating [3, 10]. The one pre-existing N-1 overload of the base case is a reliability need and is reported separately. Peak deliverability uses the model's evening hours of highest battery and non-solar output (S3) and the CAISO HSN/SSN definitions; off-peak deliverability uses the model's midday hours of highest new output after (S1) and before (S2) curtailment (Appendix Table 13). α is the largest scaling of the joint study output that meets every attributable limit.

Validation. The DC engine reproduces the earlier study's 663.31 MW and 8.62 MW results; PyPSA's PTDF, branch-outage distribution factors and linear power flow agree with it to $\leq 2 \times 10^{-9}$; LODF and multiple-outage factors agree with explicit re-solves (10^{-11} and 5×10^{-12} MW).

Results. The 2035 additions are deliverable everywhere with 5.6–7.5× margin. In 2050 the midday output is deliverable after the model's own curtailment ($\alpha = 1.22$) and almost before it (0.95), whereas evening

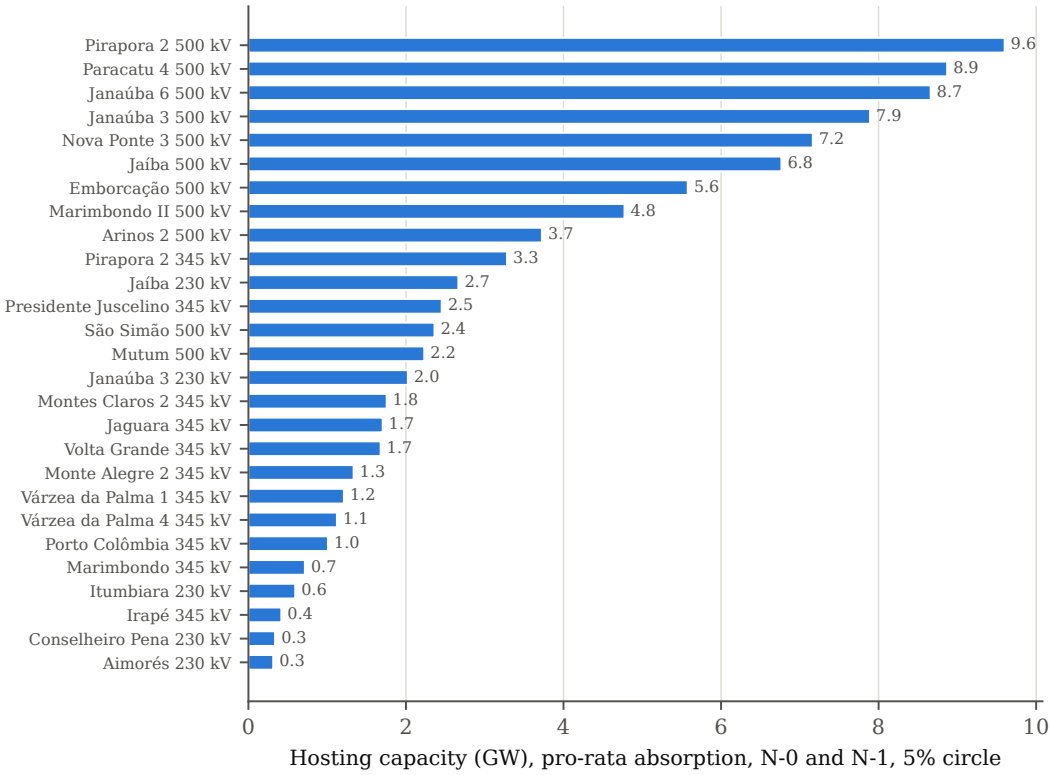


Figure 3: Per-POI hosting capacity of the 2050 candidate stations (pro-rata absorption, N-0 and N-1, 5% circle).

Table 6: Deliverability of the fixed portfolios, reference siting, main case (pro-rata absorption, target-year load, 5% circle). α: largest scaling of the joint study output meeting every attributable N-0/N-1 limit ($\alpha \geq 1$: deliverable).

Year	Study condition	Need MW	$\alpha \uparrow$	Groups ↓	New viol. ↓	Pre-ex. +MW ↓	First binding element
2035	S1 midday (LP-used)	1,552	7.48	0	0	0	Marimbondo–Porto Colômbia 345 kV
2035	S2 midday (pre-curt.)	1,585	7.34	0	0	0	Marimbondo–Porto Colômbia 345 kV
2035	S3 evening (BESS)	753	5.58	0	0	0	Marimbondo–Porto Colômbia 345 kV
2035	HSN (CAISO)	540	5.89	0	0	0	Marimbondo–Porto Colômbia 345 kV
2035	SSN (CAISO)	993	7.01	0	0	0	Marimbondo–Porto Colômbia 345 kV
2050	S1 midday (LP-used)	8,177	1.22	0	0	0	Irapé 345 kV transformer
2050	S2 midday (pre-curt.)	11,142	0.95	1	2	0	Irapé 345 kV transformer
2050	S3 evening (BESS)	7,262	0.58	8	21	91	Irapé 345 kV transformer
2050	HSN (CAISO)	7,073	0.62	9	22	91	São Gonçalo do Pará 138 kV transformer
2050	SSN (CAISO)	10,092	0.93	1	2	43	Irapé 345 kV transformer

delivery is not: $\alpha = 0.58$ (S3) and 0.62 (HSN), with 8 to 9 upgrade groups, mostly transformation at Irápe, Montes Claros 2, Janaúba 3 and São Gonçalo do Pará. The cause is the battery fleet: 3.7 of 5.3 GW are allocated to Montes Claros and Janaúba, and their full discharge flows south through northern transformers. Placing the batteries at load buses lifts evening α to 0.89 (S3) and 1.18 (HSN) (Figure 4). The portfolio also worsens the base case's N-1 overload of the São Gonçalo do Pará 500/138 kV transformation by up to 91 MW. Sensitivities (Appendix): the export-only convention scales every existing outflow tie in proportion and makes the Itajubá 3–Cachoeira Paulista interface bind first; testing every row instead of the circle drives α to zero through the pre-existing overload; the 2050 sensitivity portfolios with more storage (load +6%, dry hydro) need 14–15 groups.

P7 and AC checks (3e, 3d). Treating every pair of existing parallel circuits on one corridor as a common structure (50 pairs; conservative, tower data are not available; new circuits are on their own structures), P7 binds before P1 in the three 2050 peak cases of the reference siting (S3, HSN and SSN; Table 7): losing two of the three Itabira 2–Itabira 5 230 kV circuits overloads the third ($\alpha = 0.20$). The reinforcement plan of Section 6, sized for P1, leaves this need ($\alpha = 0.22$). Repeating the same procedure on the P7 rows adds 2 new 230 kV circuits: Itabira 2–Itabira 5 230 kV (17.6 km routed, R$40.4 M; the surviving circuit needs +21%) and then Ipatinga 1–Mesquita 230 kV (R$24.7 M): once the first circuit is in, the next binding P7 row is the plan's new Ipatinga 1–Mesquita 230 kV circuit for the loss of both existing ones. Together they cost R$65.1 M (R$6.8 M/yr), raise the reference-siting capex to R$1,314.6 M and lower the full-plan BCR to 42.3; with them

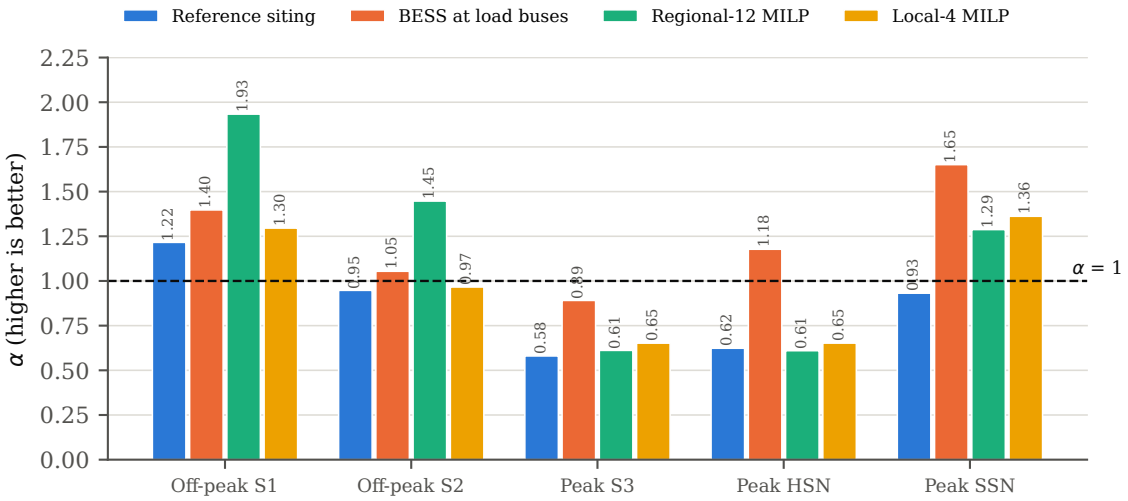


Figure 4: 2050 deliverability by study condition and siting (main case). Values below the dashed line need reinforcement.

every condition meets P1 and P7 ($\alpha \geq 1.00$). With batteries at load P7 never binds first on the existing grid, and the one-bank plan meets P1 and P7 ($\alpha \geq 1.46$). For the AC check, PyPSA's Newton–Raphson power flow with the network's injections reproduces the PWF flows (median error 0.003 MW, 93.4% of branches within 2 MW + 2%, median voltage error 0.0004 pu). The voltage screen (Table 8) finds two new high-voltage buses in the 2035 off-peak cases (S1, S2), one new low-voltage bus in the 2050 evening reference cases (S3, HSN) and, with batteries at load, a voltage rise above the normal limit at up to 27 buses in the evening as the network unloads, a condition normally handled by reactive switching that the screen does not model.

Table 7: NERC TPL P7 check (2050), α by network. P1: existing grid, P1 rows only; P1+P7: existing grid with P7 rows; Plan: verified reinforcement plan of Section 6 (sized for P1), P1+P7; Plan + remedy: with the P7 remedy circuits (— = none needed). Every pair of existing parallel circuits on one corridor is assumed to share a structure (conservative); new circuits are on their own structures. 2035 stays deliverable with P7 ($\alpha \geq 9.1$ on P7 rows).

Siting	Condition	P1 ↑	P1+P7 ↑	Plan ↑	Plan + remedy ↑	Binding P7 element (existing grid)
Reference	S1 midday (LP-used)	1.22	1.22	1.34	1.72	Itabira 2–Itabira 5 230 kV
Reference	S2 midday (pre-curt.)	0.95	0.95	1.53	1.53	Paracatu 4–Emborcação 500 kV
Reference	S3 evening (BESS)	0.58	0.20	0.22	1.00	Itabira 2–Itabira 5 230 kV
Reference	HSN (CAISO)	0.62	0.21	0.22	1.00	Itabira 2–Itabira 5 230 kV
Reference	SSN (CAISO)	0.93	0.33	0.35	1.09	Itabira 2–Itabira 5 230 kV
BESS at load	S1 midday (LP-used)	1.40	1.40	2.18	—	Arinos 2–Pirapora 2 500 kV
BESS at load	S2 midday (pre-curt.)	1.05	1.05	1.54	—	Paracatu 4–Emborcação 500 kV
BESS at load	S3 evening (BESS)	0.89	0.89	1.46	—	Irapé 345 kV transformer
BESS at load	HSN (CAISO)	1.18	1.18	1.93	—	Irapé 345 kV transformer
BESS at load	SSN (CAISO)	1.65	1.65	1.89	—	Paracatu 4–Emborcação 500 kV

Table 8: AC voltage screen (PyPSA Newton–Raphson, N-0, MG buses ≥230 kV, bus-specific normal limits of the PWF). The base case already has 5 buses above their limit. Generator reactive limits are not enforced.

Year	Siting	Condition	V_{min} (pu)	V_{max} (pu)	New low ↓	New high ↓	Min. margin (pu) ↑	Tightest bus
2035	Reference	S1 midday (LP-used)	0.878	1.091	0	2	-0.006	Itabira 5 230 kV
2035	Reference	S2 midday (pre-curt.)	0.878	1.091	0	2	-0.006	Itabira 5 230 kV
2035	Reference	S3 evening (BESS)	0.878	1.089	0	0	-0.005	Itabira 5 230 kV
2035	Reference	HSN (CAISO)	0.878	1.089	0	0	-0.004	Itabira 5 230 kV
2035	Reference	SSN (CAISO)	0.878	1.089	0	0	-0.005	Itabira 5 230 kV
2050	BESS at load	S1 midday (LP-used)	0.879	1.081	0	0	0.000	Santos Dumont 2 345 kV
2050	BESS at load	S2 midday (pre-curt.)	0.879	1.079	0	0	0.001	Santos Dumont 2 345 kV
2050	BESS at load	S3 evening (BESS)	0.879	1.100	0	26	-0.015	Itabira 5 230 kV
2050	BESS at load	HSN (CAISO)	0.879	1.101	0	27	-0.015	Itabira 5 230 kV
2050	BESS at load	SSN (CAISO)	0.879	1.085	0	0	-0.003	Santos Dumont 2 345 kV
2050	Reference	S1 midday (LP-used)	0.879	1.078	0	0	0.003	Santos Dumont 2 345 kV
2050	Reference	S2 midday (pre-curt.)	0.879	1.078	0	0	0.003	Santos Dumont 2 345 kV
2050	Reference	S3 evening (BESS)	0.879	1.073	1	0	-0.007	CVERED 345 kV
2050	Reference	HSN (CAISO)	0.879	1.073	1	0	-0.002	CVERED 345 kV
2050	Reference	SSN (CAISO)	0.879	1.075	0	0	0.008	Aurora 3 230 kV

Does a constraint subset represent the full system? The production-cost run needs a manageable set of flow constraints. Rows loaded ≥90% at full study output in any of the 220 cases form a subset of 8,393

rows (3.5% of 242,708), 66 elements in 30 groups. Because removing constraints can only make a problem more optimistic, a solution that satisfies all omitted rows is optimal for the full problem; and any row that binds at $\alpha < 1$ is loaded $\geq$100% at full output, so it is in the subset. Empirically α is identical in 106 of 106 in-sample cases with $\alpha < 1$ and in 71 of 71 out-of-sample cases (subset drawn from the reference siting only). The production-cost run therefore adds flowgates lazily and re-checks every hour on all rows (Section 7).

6 Mitigation and battery siting (3f, 2c)

For each attributable group the alternatives follow the PG&E cluster-study structure [11]: status quo; limiting-element removal or re-rating; reconductoring; transformer addition or new circuit; storage (here, siting the planned batteries at load); grid-enhancing technologies such as dynamic line rating [15]; and remedial action schemes. Costed alternatives use published unit costs (Appendix Table 20): EPE/ONS and EPE transformer and bay costs at ANEEL reference prices, EPE line costs, and EPE's finding that reconductoring costs 65–108.5% of a new circuit. Sizing assumes identical parallel elements share flow equally. The plan is then verified on a network that actually contains the new elements (PTDF and LODF rebuilt, base flows re-anchored): every study condition is deliverable on the first pass (smallest $\alpha = 1.000$).

Table 9: Attributable reinforcement plan (2050, main case), verified on a network that contains the new elements. Need: largest required increase over the study conditions. Non-wires candidate: line need $\leq$10%, where re-rating, limiting-element removal or dynamic line rating may suffice (not costable with the data). Existing: parallel elements in the group.

Siting	Upgrade group	Need (%)	Existing	Least-cost alternative	Elements	Capex (R$ M) ↓	Non-wires
Reference	Ipatinga 1–Mesquita 230 kV	3.9	2	new circuit	1	24.7	yes
Reference	Ipatinga 1–Timóteo 2 230 kV	4.3	1	reconductor (HTLS)	1	32.3	yes
Reference	Jaguara–Volta Grande 345 kV	3.1	1	reconductor (HTLS)	1	231.0	yes
Reference	Montes Claros 2–Várzea da Palma 345 kV	6.0	1	reconductor (HTLS)	1	383.4	yes
Reference	Volta Grande–L.C. Barreto 345 kV	0.4	1	reconductor (HTLS)	1	311.4	yes
Reference	Irapé 345/230 kV transf.	68.5	2	add bank(s)	2	94.1	
Reference	Janaúba 3 500/230 kV transf.	13.1	5	add bank(s)	1	57.6	
Reference	Montes Claros 2 345/138 kV transf.	23.3	2	add bank(s)	1	60.7	
Reference	São Gonçalo do Pará 500/138 kV transf.	7.5	3	add bank(s)	1	54.4	
BESS at load	Irapé 345/230 kV transf.	9.4	2	add bank(s)	1	47.0	

The reference-siting plan has 9 groups. Its costed total is R$1,250 M (R$1,009–1,490 M across the reconductoring range); five line needs are below 10% and are candidates for re-rating or dynamic rating, without which the plan is the four transformer additions, R$267 M. With batteries at load the plan is one Irápe bank, R$47 M. The pre-existing São Gonçalo do Pará overload (42.9 MW in the base case) is a reliability need requiring one bank (R$54.4 M); the portfolio's 91.1 MW raises it to 2 banks, an attributable increment of R$54.4 M (possibly the same bank as the plan's São Gonçalo do Pará addition).

7 Economic assessment (4a, 4b)

Method. The fixed portfolio is dispatched hour by hour in PyPSA with the model's costs and limits plus DC flowgates on the main grid [16]. Hourly changes relative to the network snapshot are injected on their buses: the snapshot corresponds to the 2035 model hour with the same MG load (10,731 MW) and hydro output (5,132 MW); load, existing distributed generation and unserved energy go to load buses pro rata, reservoir hydro to the hydro-plant buses, imports to the boundary inflow equivalents, existing PV equally to the PV buses, thermal to the thermal buses, and new resources and batteries to their POIs, so the changes balance exactly. Months are solved separately with the monthly hydro budgets and month-boundary battery states taken from the annual unconstrained dispatch; flowgates are added lazily until no N-0 or N-1 row is violated in any hour. Annual system cost is production cost plus unserved energy at R$8,327.76/MWh [4] plus unused zero-cost energy (curtailment of the profile-based resources as defined in the model, and unused hydro budget) at the 2025 SE marginal cost, R$209.47/MWh (the model's import price). Congestion curtailment is the unused zero-cost energy above the same year's run without network limits, as a share of the available profile-based and hydro energy (the split between the two is not unique); we screen it against a 5% threshold (a study choice). Without flowgates the monthly runs reproduce the annual dispatch (no unserved energy; same imports and production cost).

Results. In 2035 the network constrains only 2 rows briefly and adds no congestion cost (R$0 M/yr). In 2050 the existing grid sheds 689.5 GWh of load (0.71%) in every month, a congestion cost of R$5,824 M/yr that is almost entirely unserved energy (Figure 5); without network limits the same dispatch serves all load. The reinforcement plan reduces unserved energy to 5.9 GWh, all in July, the month with the smallest hydro

Table 10: Network-constrained 8,760-h production cost (PyPSA, monthly, lazy flowgates on all N-0/N-1 rows). System cost = production + unserved energy at R$ 8,327.76/MWh + unused zero-cost energy (curtailment of profile-based resources and unused hydro budget) at R$ 209.47/MWh. Unused %: share of the available profile-based and hydro energy; Cong. curt.: unused energy above the same year without network limits, in the same units.

Scenario	Production ↓	Unserved GWh ↓	Unused TWh ↓	Unused % ↓	Cong. curt. % ↓	System cost ↓	Congestion ↓	Rows
2035, no network limits	480.2	0.0	1.47	1.8	0.00	787	0	0
2035, existing grid	480.2	0.0	1.47	1.8	0.00	787	0	2
2050, no network limits	44.6	0.0	15.01	13.4	0.00	3,189	0	0
2050, existing grid	48.3	689.5	15.38	13.7	0.33	9,013	5,824	679
2050, grid + plan	46.6	5.9	15.03	13.4	0.01	3,243	54	401
2050 BESS at load, existing grid	45.2	78.3	15.08	13.4	0.06	3,857	667	319
2050 BESS at load, grid + plan	44.7	46.7	15.06	13.4	0.04	3,588	399	298

Costs in R$ M/yr; Rows: flowgate rows added by the lazy check.

energy budget (3,058 GWh against 3,478–4,684 GWh in the other months), and its benefit-cost ratio is 44.5 for the full costed plan and 208.3 if small line needs are met by non-wires means (Table 11); because the benefit is mostly avoided unserved energy, it stays far above one in every valuation sensitivity (minimum 22.5). Siting the batteries at load is worth R$5,156 M/yr before any grid investment: unserved energy falls to 78.3 GWh, again only in July. Its one-bank plan leaves 46.7 GWh in July; the flowgates that the lazy check adds concentrate on Timóteo–Timóteo 2 230 kV (181 rows), São Gonçalo do Pará 138 kV transformer (53) and Santos Dumont 2 500/345 kV (51). Apart from the pre-existing São Gonçalo do Pará overload, the study-condition screen did not flag these elements; they are candidates for an economic-driven addition. The energy-tight month, not the study hours, sets the residual need, which is why the production-cost run is part of the assessment.

What drives curtailment? Not the grid. Without any network limit, 15.01 TWh of zero-cost energy (13.4% of the available profile-based and hydro energy) goes unused in 2050 because the hourly local-balance rule allows no exports; the existing grid adds congestion curtailment of only 0.33% (0.01% with the plan), well below the 5% screen. Unlike NREL's Oregon finding that congestion drives curtailment [18], the grid's cost here is unserved load, not lost energy, and the curtailed energy is what an export option would sell (next paragraph).

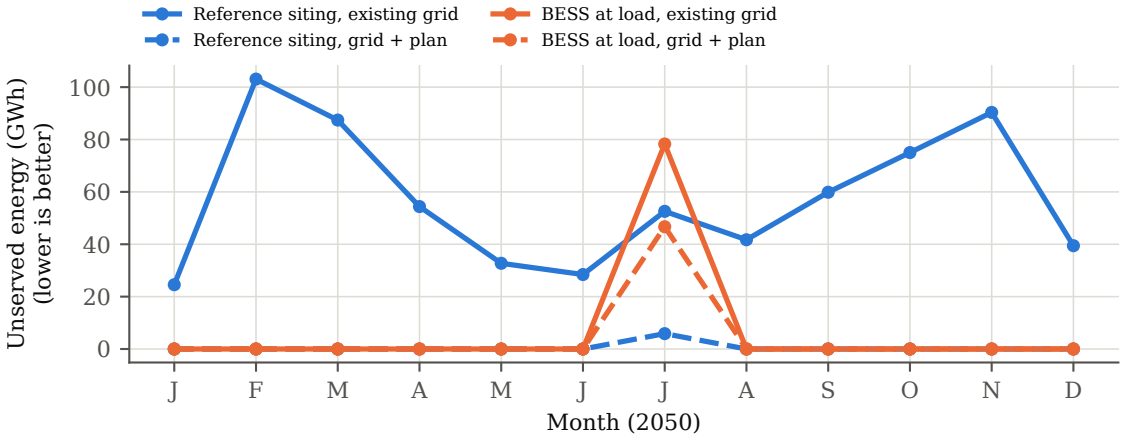


Figure 5: Monthly unserved energy in the 2050 network-constrained production-cost runs.

Table 11: Benefit-cost ratios (annual benefit / annual revenue requirement at CRF(8%, 30 yr) + 1.5% O&M). BCR > 1 justifies the measure (CAISO/PJM economic criterion). Valuing unused zero-cost energy ±50% changes the BCRs by at most 1.0%.

Measure	Valuation	Benefit (R$ M/yr) ↑	Cost, full plan (R$ M/yr)	BCR ↑	Cost, excl. small needs	BCR ↑
Plan, reference siting	base	5,770	129.7	44.5	27.7	208.3
Plan, BESS at load	base	269	4.9	55.0	4.9	55.0
Plan, reference siting	VOLL -50%	2,923	129.7	22.5	27.7	105.5
Plan, BESS at load	VOLL -50%	137	4.9	28.1	4.9	28.1
Plan, reference siting	VOLL +50%	8,617	129.7	66.4	27.7	311.1
Plan, BESS at load	VOLL +50%	400	4.9	82.0	4.9	82.0

Hourly or annual matching. The 2050 requirement is hourly local balance. Allowing exports alone lowers cost by 13.4% (16.0 TWh exported); annual matching (imports allowed hour by hour, annual imports no greater than exports) lowers it by 48.7% and removes the need for batteries, wind and biogas (Table 12), in line with the course's 24/7 case [19]. That alternative relies on 16.6 TWh/yr of exports, which the export interface results above would have to accommodate.

Table 12: Hourly versus annual clean matching in 2050 (capacities in GW; gas is zero in every case).

2050 definition	Cost (R$ M PV) ↓	vs H0 (%) ↓	Central PV	Wind	BESS	Imports (TWh)	Exports (TWh)
H0 hourly local balance (LP)	20,285	0.0	12.92	2.00	5.29	0.0	0.0
H1 hourly balance, exports allowed	17,557	-13.4	14.11	2.00	4.83	0.0	16.0
A1 annual matching (net annual exporter)	10,415	-48.7	10.75	0.00	0.00	12.0	16.6

8 Findings

1. **2035: no major grid improvement is needed.** The policy portfolio is deliverable in every study condition with 5.6–7.5× margin, adds no congestion cost, and is adequate under the model's import assumption.
2. **2050 midday solar is deliverable; 2050 evening storage discharge is not.** Midday $\alpha = 1.22$; evening $\alpha = 0.58$–0.62 because 3.7 GW of batteries sit in the north.
3. **Battery siting is the cheapest fix.** At load buses the evening need falls to one transformer bank (R$47 M) and the system cost falls by R$5,156 M/yr.
4. **If batteries stay in the north, the attributable plan is justified.** 9 groups, R$267–1,250 M, BCR 44.5–208.3. If existing parallel circuits share towers (P7), 2 more 230 kV circuits (R$65.1 M) are needed and the full-plan BCR is 42.3.
5. **2050 adequacy is energy-limited.** A capacity-only COPT finds no shortfall; the energy-limited simulation finds shortfalls in every simulated year and 2,859.3 GWh in a dry year, needing up to 94 MW (normal) and 1,594 MW (dry) of firm capacity.
6. **Curtailment comes from the hourly local-balance rule, not from congestion.** Even without network limits 15.01 TWh (13.4%) of 2050 zero-cost energy goes unused; congestion adds 0.33% (below the 5% screen).
7. **The definition of fossil-free matters.** Annual instead of hourly matching lowers 2050 cost by 48.7%, if exports can be delivered.

Limitations. One network snapshot (evening) and one weather year; the 2050 network is the 2035 topology; DC flows except for the N-0 AC voltage screen, which covers the existing grid only and does not enforce generator reactive limits; hydro unit sizes are a proxy; reconductoring is modelled as a rating increase with unchanged impedance and new circuits and banks as added parallel elements; P7 assumes that existing parallel circuits share structures; the production-cost benefits are computed for the P1 plan and held at their 2050 value over the asset life.

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A Supplementary tables

Table 13: Study output levels (fraction of new nameplate). S1–S3: mean of the top 1% LP hours; HSN/SSN: CAISO exceedance definitions.

Year	Technology	Off-peak S1	Off-peak S2	Peak S3	Peak HSN	Peak SSN
2035	bess	0.00	0.00	0.00	1.00	0.50
2035	bio	0.33	0.34	0.46	0.43	0.36
2035	biogas	0.00	0.00	0.00	0.70	0.70
2035	small hydro	0.36	0.36	0.33	0.54	0.43
2035	solar central	0.58	0.59	0.14	0.00	0.36
2035	solar distributed	0.55	0.56	0.11	0.00	0.18
2035	wind site	0.00	0.00	0.00	0.37	0.24
2050	bess	0.11	0.11	1.00	1.00	0.50
2050	bio	0.16	0.20	0.45	0.43	0.36
2050	biogas	0.09	0.09	0.58	0.70	0.70
2050	small hydro	0.35	0.46	0.34	0.54	0.43
2050	solar central	0.31	0.44	0.01	0.00	0.36
2050	solar distributed	0.31	0.43	0.00	0.00	0.18
2050	wind site	0.32	0.42	0.49	0.37	0.24

Table 14: Sensitivity of $\alpha \uparrow$ to the absorption convention (reference siting, 5% circle). For 2050 the main case includes target-year load growth; "2035 load" omits it.

Year	Condition	Pro rata	A1 MG only	A2 inflows	B export	Pro rata, 2035 load	B, 2035 load
2035	Off-peak S1 (midday, LP-used)	7.48	3.41	7.14	4.96	—	—
2035	Off-peak S2 (midday, pre-curtailment)	7.34	3.34	7.00	4.86	—	—
2035	Peak S3 (evening, BESS)	5.58	6.02	5.49	5.11	—	—
2035	Peak HSN (CAISO)	5.89	6.23	5.81	5.50	—	—
2035	Peak SSN (CAISO)	7.01	5.23	6.81	6.08	—	—
2050	Off-peak S1 (midday, LP-used)	1.22	0.68	1.13	1.24	1.24	0.91
2050	Off-peak S2 (midday, pre-curtailment)	0.95	0.46	0.87	0.85	0.96	0.67
2050	Peak S3 (evening, BESS)	0.58	0.45	0.56	0.59	0.59	0.59
2050	Peak HSN (CAISO)	0.62	0.45	0.63	0.66	0.66	0.67
2050	Peak SSN (CAISO)	0.93	0.46	0.87	0.95	0.95	0.76

Table 15: Sensitivity of $\alpha \uparrow$ to the attribution rule. "Every row tested" also forbids any worsening of the one pre-existing overload, which drives α to zero; the cap variant raises only the highest-impact circle units (a bound).

Year	Condition	Every row tested	5% circle (main)	Circle + 20 unit / 1,500 MW cap
2035	Off-peak S1 (midday, LP-used)	7.48	7.48	7.62
2035	Off-peak S2 (midday, pre-curtailment)	7.34	7.34	7.48
2035	Peak S3 (evening, BESS)	5.58	5.58	5.60
2035	Peak HSN (CAISO)	0.00	5.89	5.88
2035	Peak SSN (CAISO)	7.01	7.01	7.03
2050	Off-peak S1 (midday, LP-used)	0.00	1.22	1.16
2050	Off-peak S2 (midday, pre-curtailment)	0.95	0.95	0.89
2050	Peak S3 (evening, BESS)	0.00	0.58	0.85
2050	Peak HSN (CAISO)	0.00	0.62	1.11
2050	Peak SSN (CAISO)	0.00	0.93	0.89

Table 16: Sensitivity of 2050 $\alpha \uparrow$ to siting (main case).

Condition (2050)	Reference	Reference, BESS at load	Regional-12 MILP	Local-4 MILP
Off-peak S1 (midday, LP-used)	1.22	1.40	1.93	1.30
Off-peak S2 (midday, pre-curtailment)	0.95	1.05	1.45	0.97
Peak S3 (evening, BESS)	0.58	0.89	0.61	0.65
Peak HSN (CAISO)	0.62	1.18	0.61	0.65
Peak SSN (CAISO)	0.93	1.65	1.29	1.36

Table 17: Midday sensitivity with existing central PV (split equally over the 106 UFV buses) and existing DG added.

Year	Absorption	Need (MW)	$\alpha \uparrow$	Status
2035	Pro rata	9,666	2.39	deliverable
2035	A1 MG only	9,666	0.51	energy-limited
2035	A2 inflows	9,666	2.26	deliverable
2035	B export	9,666	0.77	needs upgrades
2050	Pro rata	12,743	1.20	deliverable
2050	A1 MG only	12,743	0.42	energy-limited
2050	A2 inflows	12,743	1.06	deliverable
2050	B export	12,743	0.71	needs upgrades

Table 18: Peak deliverability of the 2050 sensitivity portfolios (reference siting by allocation weights, main case). The 2035 sensitivity portfolios equal the base portfolio (policy floors bind) and stay deliverable ($\alpha \geq 3.45$).

Sensitivity portfolio (2050)	Cond.	Need (MW)	$\alpha \uparrow$	Upgrade groups $\downarrow$
load +6%	HSN	8,364	0.46	15
load +6%	SSN	12,352	0.74	5
load -6%	HSN	5,140	0.83	1
load -6%	SSN	8,346	1.15	0
hydro dry	HSN	10,634	0.47	14
hydro dry	SSN	12,648	0.75	3
hydro wet (param.)	HSN	5,525	0.63	4
hydro wet (param.)	SSN	8,538	1.10	0
gas +50%	HSN	7,073	0.62	9
gas +50%	SSN	10,092	0.93	1
gas -25%	HSN	7,073	0.62	9
gas -25%	SSN	10,092	0.93	1

Table 19: Effective load-carrying capability by the CAISO calibration (Lecture 11, method A).

Year	Counting	Resource class	Nameplate (MW)	ELCC (MW) $\uparrow$	ELCC (% of nameplate) $\uparrow$
2035	nameplate	new solar (central + DG)	2,044	1	0.0
2035	nameplate	new biomass + small hydro	1,173	305	26.0
2035	nameplate	all new resources	3,217	306	9.5
2050	nameplate	new solar (central + DG)	20,862	3,304	15.8
2050	nameplate	new BESS	5,290	4,734	89.5
2050	nameplate	new wind	2,000	391	19.5
2050	nameplate	new biomass + small hydro	1,548	523	33.8
2050	nameplate	new biogas	400	256	64.1
2050	nameplate	all new resources	30,101	5,786	19.2
2050	deliverable only	new solar (central + DG)	20,862	928	4.4
2050	deliverable only	new BESS	5,290	3,082	58.2
2050	deliverable only	new wind	2,000	215	10.7
2050	deliverable only	new biomass + small hydro	1,548	323	20.9
2050	deliverable only	new biogas	400	156	38.9
2050	deliverable only	all new resources	30,101	3,786	12.6

Table 20: Unit costs used for the costed alternatives (published EPE/ANEEL values; price bases differ and are not indexed).

Item	Unit	Value (R$ M or ratio)	Source	Price basis
ATF single-phase unit 500/345 kV 500 MVA	per unit (500 MVA)	23.875	EPE/ONS EPE-DEE-NT-047-2022 (RT-ONS DPL 0355/2022) Avaliação dos benefícios sistêmicos do TR 500/345 kV na SE UTE GNA	ANEEL BPR (as used in the note)
ATF single-phase unit 500/230 kV 100 MVA	per unit (100 MVA)	10.975	EPE-DEE-RE-071-2025-rev0	ANEEL Ref. 2025/0123
ATF single-phase unit 345/138 kV 100 MVA	per unit (100 MVA)	11.232	EPE-DEE-NT-029-2023-rev0 Ampliação da SE Campinas 345/138 kV	ANEEL BPR base March 2023
CT transformer connection bay 500 kV (DJM)	per bay	15.044	EPE-DEE-NT-047-2022	ANEEL BPR (as used in the note)
CT transformer connection bay 345 kV (DJM)	per bay	12.106	EPE-DEE-NT-047-2022	ANEEL BPR (as used in the note)
CT transformer connection bay 230 kV (BD4)	per bay	9.668	EPE-DEE-RE-071-2025-rev0	ANEEL Ref. 2025/0123
CT transformer connection bay 138 kV (BD4)	per bay	6.430	EPE-DEE-NT-029-2023-rev0	ANEEL BPR base March 2023
New line 230 kV	per km	1.104	EPE-DEE-RE-071-2025-rev0 (inputs/cost_assumptions.json)	EPE 2025 (project cost file)
New line 345 kV single circuit 4x954 MCM	per km	2.388	EPE-DEE-NT-047-2022	ANEEL BPR (as used in the note)
New line 500 kV	per km	2.469	EPE-DEE-RE-071-2025-rev0 (inputs/cost_assumptions.json)	EPE 2025 (project cost file)
Line entry bay 230 kV	per bay	10.501	EPE-DEE-RE-071-2025-rev0 (inputs/cost_assumptions.json)	EPE 2025 (project cost file)
Line entry bay 345 kV (DJM)	per bay	10.518	EPE-DEE-NT-047-2022	ANEEL BPR (as used in the note)
Line entry bay 500 kV	per bay	13.398	EPE-DEE-RE-071-2025-rev0 (inputs/cost_assumptions.json)	EPE 2025 (project cost file)
Reconductoring vs new line (total cost ratio)	ratio	0.650	EPE-DEE-RE-071-2025-rev0	ANEEL Ref. 2025/0123
Reconductoring vs new line (total cost ratio)	ratio	1.085	EPE-DEE-RE-071-2025-rev0	ANEEL Ref. 2025/0123

B Reproducibility

All results are produced by scripts in the repository (`hosting_capacity/`, `planning_study/`); every table and figure and most numbers in the text are generated from their CSV outputs (`planning_study/report/make_report.py`), and the few values typed into the text were checked against them. The environment is listed in `planning_study/requirements.txt` (PyPSA 1.2.4, linopy 0.10.1, HiGHS 1.16, pandas 2.3.3). Third-party input data are not redistributed. Readable station names are inferred from network bus codes.